

Learning classification on hierarchical ultrametric domains: generalization bounds in the p -adic setting

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Abstract

Many data domains are naturally hierarchical, with similarity governed by taxonomy-like or tree-structured relations rather than by Euclidean geometry. Ultrametric spaces provide a principled mathematical model for such settings, but their role in statistical learning theory remains comparatively underdeveloped. In this paper, we study binary classification on the ultrametric domain of p -adic integers, viewed as a canonical model of hierarchical structure. We consider concept classes given by finite unions of p -adic balls and introduce a signed-width representation that places these concepts within a margin-based learning framework. This formulation yields a tractable analysis of capacity and generalization: we show that the associated real-valued function class has pseudo-dimension 1, derive covering-number bounds, and obtain non-asymptotic generalization guarantees for large-width classification. The results highlight how the geometry of ultrametric domains can lead to remarkably simple complexity behavior, despite the richness of the underlying concept classes. Beyond the specific p -adic setting, the paper provides a mathematically explicit framework for studying classification and generalization on hierarchical domains.

1 Introduction

Many learning problems involve data whose similarity structure is inherently hierarchical rather than Euclidean. Examples include phylogenetic data, taxonomies, ontologies, directory trees, and other domains in which proximity is governed by ancestry, refinement, or common descent. In such settings, the relevant geometry is often better captured by an ultrametric than by a Euclidean metric. This motivates the study of classification and generalization in ultrametric domains.

Ultrametric spaces are metric spaces in which the distance satisfies the strong triangle inequality: for any three points x, y, z ,

$$d(x, y) \leq \max \{d(x, z), d(z, y)\}. \quad (1)$$

Their geometry is naturally hierarchical: balls are nested, and triangles are isosceles with the two longer sides equal. As a result, ultrametric spaces provide a principled mathematical model for hierarchical similarity.

Examples arise in phylogenetic trees [1, 2], p -adic analysis [3], and computer vision [4]. A rooted tree or dendrogram induces an ultrametric, and conversely the family of balls in an ultrametric space reflects a hierarchy. Hierarchical data often appears in the form of a taxonomy; one example is WordNet [5], whose synonym sets are arranged through semantic relations.

Preprint timestamps of all versions of the paper:

<https://github.com/ratsaby/joel-ratsaby-preprints/tree/main/p-adic-concepts>

Distances induced by taxonomies are not always represented well by Euclidean embeddings. This has led to growing interest in methods that use hierarchical distance information directly. Recent work includes neural architectures in ultrametric p -adic space [6], low-distortion ultrametric embeddings of Euclidean data [7], learning ultrametric trees for optimal transport regression [8], incremental treatment of tree metrics and ultrametrics in data streams [9], and ultrametric feature spaces for hierarchical segmentation [4]. Despite this growing body of work, the learning-theoretic analysis of classification in ultrametric domains remains relatively limited.

In statistical learning theory, one basic principle is that classifiers that separate oppositely labeled data with a large margin tend to generalize better [10–14]. A related notion is that of learning large *width* classifiers, introduced in [15], extended in [16–25] to various kinds of input domains and classes of functions, and implemented as a learning algorithm [26]. The large-width framework provides a natural way to quantify separation in metric spaces and is therefore well suited to hierarchical settings.

Consider a sample of labeled points $\{(x_i, y_i)\}_{i=1}^m$, with $x_i \in X$, classification labels $y_i \in Y := \{-1, 1\}$, $1 \leq i \leq m$, and an input space X with distance function $d : X \times X \rightarrow \mathbb{R}_{\geq 0}$ where $\mathbb{R}_{\geq 0}$ denotes the set of non-negative real numbers. A *classifier* is a function $h : X \rightarrow Y$. Let S_h^+ be a subset of X classified as 1 and S_h^- a subset classified as -1 , by h . If x is a sample point with a label $y = 1$ then we define the classification width at x to be the distance $d(x, S_h^-)$ defined as the minimum (or infimum, in the case that S_h^- is infinite) of the distances between x and elements of S_h^- . Similarly, if x is labeled -1 then the width at x is defined as the distance $d(x, S_h^+)$. The sample width is defined to be the minimum, over the sample points, of these individual widths.

The generalization error bounds obtained in [15–25] improve as the sample width increases. This motivates the study of large-width classification as a framework for understanding generalization in structured metric domains, including ultrametric ones.

In this paper we study binary classification on the ultrametric domain $\mathbb{Z}_p$ of p -adic integers, where p is a prime number. The space $\mathbb{Z}_p$ is infinite, compact, and admits a natural rooted-tree representation in which each node has a unique parent, except the root, while p -adic integers may be viewed as leaves at infinite depth, that is, boundary points of the infinite tree [27]. It therefore provides a canonical and mathematically tractable model of hierarchical structure.

The space $\mathbb{Z}_p$ is of course a standard object in p -adic analysis and number theory [3, 27, 28]. For the purposes of this paper, however, its importance lies in the fact that it is equipped with the p -adic metric, with respect to which it is an ultrametric space, and that its balls admit an explicit description as residue classes modulo powers of p . An element of $\mathbb{Z}_p$ may be viewed as an infinite sequence of digits in $\{0, \dots, p-1\}$, and a ball is determined by a finite prefix of this sequence. This makes $\mathbb{Z}_p$ a convenient concrete model for hierarchical domains, in which geometric notions such as distance and width can be expressed through explicit symbolic and arithmetic descriptions of concepts.

We consider classifiers based on concepts that are finite unions of maximal balls in $\mathbb{Z}_p$. To analyze them, we use a signed-width representation that places these concepts within a margin-based real-valued framework. This leads to a tractable analysis of capacity and generalization for classification on a hierarchical ultrametric domain.

1.1 Contribution

The contributions of the paper are as follows. First, we formalize a concept class on $\mathbb{Z}_p$ consisting of finite unions of maximal p -adic balls. Since balls in $\mathbb{Z}_p$ correspond to residue classes modulo powers of p , this yields an explicit and mathematically tractable model for hierarchical classification in an ultrametric domain.

Second, we introduce a signed-width representation for these concepts. The representation associates with each classifier a real-valued signed-width function whose sign recovers the classifier itself. This embeds the classification problem into a margin-based learning framework and makes it possible to analyze

generalization through notions of width and margin, extending the large-width approach of [15, 25] to the p -adic setting. In this representation, the width at a point is determined by the hierarchical position of the corresponding ball in the tree: larger width corresponds to coarser balls and therefore to simpler residue-class descriptions of the concept.

Third, we derive upper bounds on the complexity of the induced function classes. A key observation is that the ultrametric structure of $\mathbb{Z}_p$ imposes a strong combinatorial regularity on balls: any two balls are either nested or disjoint. Using this property, we show that the combinatorial dimension of $\mathbb{Z}_p$ equals 1, which in turn implies that the associated class of distance functions has pseudo-dimension 1. This yields especially simple covering-number bounds compared to those arising in more general metric settings.

Fourth, we develop a covering-number analysis for the class of signed-width functions associated with unions of at most n balls. The proof combines the low pseudo-dimension of the underlying distance class with a combinatorial counting argument based on dendrogram partitions of finite samples. More specifically, intersections of p -adic balls with a finite sample induce a hierarchical clustering structure, and the number of admissible partitions generated by these clusters controls the size of the approximating family. This leads to explicit covering-number bounds for the signed-width class.

Finally, we apply these complexity bounds to obtain non-asymptotic generalization guarantees for large-width classification on $\mathbb{Z}_p$. The resulting bounds hold uniformly over the width parameter and show that, for classifiers consistent with the sample, the error bound improves as the sample width increases. Thus, beyond the specific p -adic setting, the paper provides a mathematically explicit framework for studying classification and generalization on hierarchical ultrametric domains.

The paper is organized as follows: Section 2 reviews the required background on ultrametric spaces and the p -adic integers $\mathbb{Z}_p$, including their tree representation, ultrametric geometry, and the structure of p -adic balls. Section 3 introduces the class of p -adic concepts defined as finite unions of maximal balls and the associated binary classifiers. Section 4 develops the notion of classification width and signed width in the p -adic setting, establishes their connection to the hierarchical structure of the underlying concepts, and studies maximal-width classifiers on finite samples. Section 5 analyzes the combinatorial complexity of the induced function classes, deriving bounds on combinatorial dimension, pseudo-dimension, and covering numbers. Section 6 constructs approximating families for the signed-width function class and obtains covering-number bounds for these classes. Finally, Section 7 applies these results to statistical learning theory, deriving non-asymptotic large-width generalization bounds for classification on $\mathbb{Z}_p$, while Section 8 concludes with a discussion of the implications of the results and directions for future research.

2 Background

2.1 Ultrametric space

Let (X, d) be a metric space.

Definition 1. (*Ultrametric Space*) A metric space (X, d) is called an *ultrametric* space if for any $x, y, z \in X$, $d(x, y) \leq \max \{d(x, z), d(z, y)\}$. This inequality is called the *strong triangle inequality*.

Proposition 2. (*Isosceles property*) Let (X, d) be an ultrametric space. For any $x, y, z \in X$, if $d(x, z) < d(y, z)$, then $d(x, y) = d(y, z)$.

Proof. By the strong triangle inequality,

$$d(x, y) \leq \max \{d(x, z), d(z, y)\} = d(y, z). \quad (2)$$

On the other hand, $d(y, z) \leq \max \{d(y, x), d(x, z)\}$. Since $d(x, z) < d(y, z)$, the maximum on the right cannot be $d(x, z)$. Therefore it must be $d(y, x)$, and hence $d(y, z) \leq d(y, x) = d(x, y)$. Combining the latter with (2) gives $d(x, y) = d(y, z)$. $\square$

This is called the Isosceles property because in a triangle with vertices x, y, z , the two longer sides satisfy $d(x, y) = d(y, z)$.

A ball in X is defined as the set of all points whose distance from a center point is no more than some constant, which is referred to as its radius. An open ball of radius r centered at $a \in X$ consists of all points $z \in X$ whose distance from a is less than r . A closed ball of radius r centered at $a \in X$ consists of all points $z \in X$ whose distance from a is less than or equal to r .

Proposition 3. [3, Proposition 18.5] *Let B_1 and B_2 be balls in an ultrametric space X . Then either B_1 and B_2 are ordered by inclusion, $B_1 \subseteq B_2$ or $B_2 \subseteq B_1$, or B_1 and B_2 are disjoint.*

Proposition 4. [3] *Let B be a ball in an ultrametric space X . Then B is both open and closed.*

2.2 $\mathbb{Z}_p$

Henceforth, p denotes a prime number. The space $\mathbb{Z}_p$ of p -adic integers [27] consists of all integers x that are represented as infinite expansions

$$x = a_0 + a_1p + a_2p^2 + \cdots \quad (3)$$

where the expansion coefficients $a_i \in \mathcal{I}_p$ and

$$\mathcal{I}_p := \{0, 1, \dots, p-1\}.$$

The space $\mathbb{Z}_p$ can be alternatively viewed [3] as a space of infinite sequences

$$(a_0, a_1, \dots) \quad (4)$$

of coefficients of p -adic integers (3). $\mathbb{Z}_p$ forms a ring under the standard p -adic addition and multiplication operations.

If for a finite integer $N \geq 0$ the coefficients in the expansion of x , satisfy $a_i = 0$, for all $i \geq N$, then x is a non-negative integer (a natural number). Therefore the set $\mathbb{N}$ of natural numbers is a subset of the set of p -adic integers $\mathbb{N} \subset \mathbb{Z}_p$. A negative integer has a tail $a_i = p-1$, $i \geq M$, starting from some finite M .

For a p -adic integer $x \in \mathbb{Z}_p$ let $\nu_p(x)$ denote its *valuation*, or *order* [3, Definition 3.4], which is defined as a non-negative integer k such that $x = p^k c$ where $p \nmid c$. The valuation of a p -adic integer x is zero if p does not divide x . Equivalently, the valuation can be expressed as

$$\nu_p(x) = \begin{cases} \max \{k : p^k \mid x\} & \text{if } x \neq 0 \\ \infty & \text{if } x = 0. \end{cases}$$

The valuation measures the unique exponent of p dividing x .

The p -adic absolute value of x is defined by

$$|x|_p := \frac{1}{p^{\nu_p(x)}}$$

where $|0|_p = 1/p^\infty = 0$. It follows that for all $x \in \mathbb{Z}_p$,

$$|x|_p \leq 1 \quad (5)$$

because $\nu_p(x) \geq 0$.

Definition 5. [29, p. 9] (*Distance*) For $x, z \in \mathbb{Z}_p$, define the p -adic distance by

$$d(x, z) := |x - z|_p. \quad (6)$$

The distance (6) satisfies the strong triangle inequality (1) hence it is an ultrametric. Therefore we have the following proposition.

Proposition 6. *The space $\mathbb{Z}_p$ equipped with the p -adic distance (6) is an ultrametric space.*

Corollary 7. *The strong triangle inequality (1) and the Isosceles property (Proposition 2) hold for $\mathbb{Z}_p$.*

The tree $\mathbb{T}_p$ corresponding to $\mathbb{Z}_p$ is defined as follows.

Definition 8. (*Tree*) An infinite tree $\mathbb{T}_p$ corresponds to $\mathbb{Z}_p$. It has a single root at level -1 , and nodes at infinitely many levels $k \in \{0, 1, \dots\}$. At the k^{th} level there are p^{k+1} nodes $a = [a_0, a_1, \dots, a_k]$, where $a_i \in \mathbb{I}_p$, $0 \leq i \leq k$, and the nodes are divided into p^k disjoint groups of nodes. Each group consists of p *sibling* nodes a that share k *common ancestor* nodes (excluding the root) $[a_0], [a_0, a_1], \dots, [a_0, a_1, \dots, a_{k-1}]$, where the deepest common ancestor $[a_0, a_1, \dots, a_{k-1}]$ is referred to as *parent* and is denoted $\pi(a) = [a_0, a_1, \dots, a_{k-1}]$. The p -adic integers $x = (x_0, x_1, \dots)$ correspond to the *leaves* of $\mathbb{T}_p$. A node (or leaf) is called *descendant* of a node a if a is one of its ancestors. Every node except the root has a unique parent and every node is connected to the root by a unique finite path. Hence $\mathbb{T}_p$ is a rooted tree.

Definition 9. (*Level*) We denote by $\ell(a) \geq 0$ the *level* of a node a in $\mathbb{T}_p$. The level of the root is defined to be -1 .

Figure 1 displays an infinite tree $\mathbb{T}_3$ representation of $\mathbb{Z}_3$ where 3-adic integers are represented by leaves (not seen since they are infinitely deep). Given two 3-adic integers $x = (x_0, x_1, \dots)$ and $z = (z_0, z_1, \dots)$, suppose for example that $x_0 = z_0 = a_0$ and $x_1 \neq z_1$, then the deepest common ancestor is their parent node $[a_0]$ at level 0. Hence the valuation of their difference $\nu_3(x - z) = 1$ and therefore the distance $|x - z|_3 = 1/3$. Two unique nodes at level 1 correspond to two disjoint balls B and B' . Any p -adic integers $x, z \in B$ and $x' \in B'$ satisfy $\nu_3(x - z) = 2$ and $\nu_3(x - x') \leq 1$.

In general, at level $k \in \{0, 1, \dots\}$, a node has p children in level $k + 1$. Each such child corresponds to a choice of the next digit $a_{k+1} \in \{0, \dots, p - 1\}$ in the p -adic expansion of an integer. At level k , every node of $\mathbb{T}_p$ is a vector

$$[a_0, a_1, \dots, a_k] \in \mathcal{I}_p^{k+1} \quad (7)$$

which represents a *residue class* (or a coset)

$$\sum_{i=0}^k a_i p^i + p^{k+1} \mathbb{Z}_p = \left\{ x \in \mathbb{Z}_p : x \equiv \sum_{i=0}^k a_i p^i \pmod{p^{k+1}} \right\}. \quad (8)$$

This residue class is a closed *ball* of radius $1/p^{k+1}$ that consists of all p -adic integers whose p -adic expansion starts with $a_0, a_1, \dots, a_k$.

Note that $[a_0, a_1, \dots, a_k]$ is *not* a p -adic integer since it is not an infinite sequence of the form (4). If we append a trail of zeros to it then it is a p -adic integer (and also, a natural integer).

That Proposition 3 holds for $\mathbb{Z}_p$ follows from the fact that two balls satisfy $B \subseteq B'$ if and only if the corresponding node a' of B' is an ancestor of the corresponding node a of B in the tree $\mathbb{T}_p$, otherwise they are disjoint.

Definition 10. (*LCA*) For x and z in $\mathbb{Z}_p$, the *Least Common Ancestor* node $LCA(x, z)$ is their deepest (maximum level) common ancestor.

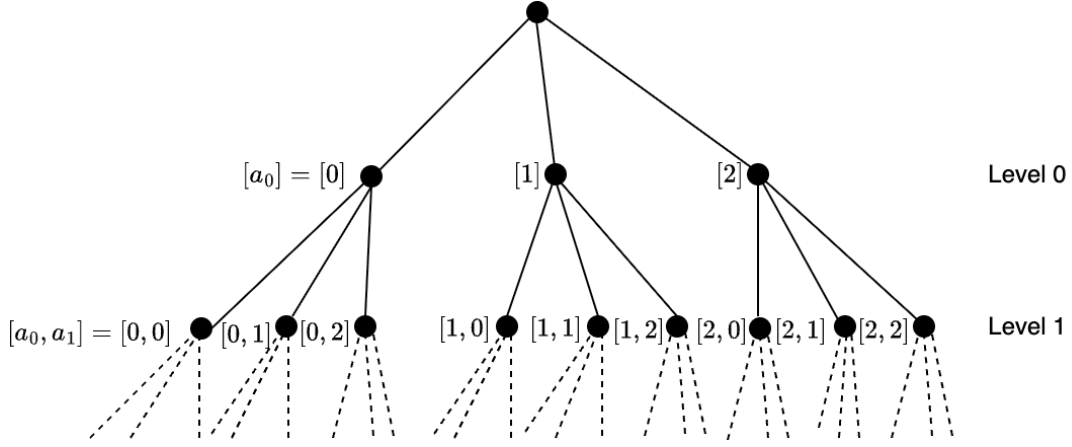


Figure 1: Tree T_p corresponding to $\mathbb{Z}_p$ ($p = 3$), has infinite number of levels $k \in \{0, 1, \dots\}$. The dashed lines connect to the next level (not shown). Leaves correspond to the p -adic integers and are not shown since their level $k = \infty$.

The word *least* is with respect to the partial order of descendency, where descendant $\leq$ ancestor.

Proposition 11. *The distance (6) between two p -adic integers x and z satisfies*

$$d(x, z) = 1/p^{\ell(LCA(x, z)) + 1}. \quad (9)$$

Definition 12. (*Diameter*) For a subset A of a metric space (X, d) , its diameter is defined by $\text{diam}(A) := \sup\{d(x, y) : x, y \in A\}$.

For any $x, z \in \mathbb{Z}_p$,

$$\ell(LCA(x, z)) \geq -1$$

because by Definition 9, the root has level -1 . Therefore from (9), we have

$$\begin{aligned} 1/p^{\ell(LCA(x, z)) + 1} &\leq 1/p^{-1+1} \\ &= 1. \end{aligned}$$

Consider for example $x = (0, 0, \dots)$ and $x = (1, 1, \dots)$. At every level $l > -1$, they do not share a common ancestor, hence $\ell(LCA(x, z)) = -1$. Hence there exists a pair whose distance is 1, and therefore

$$\text{diam}(\mathbb{Z}_p) = 1. \quad (10)$$

Definition 13. (*LCA*) For a node $a \in T_p$ and $x \in \mathbb{Z}_p$, the Least Common Ancestor node is denoted by $LCA(x, a)$. If a is an ancestor of x then $LCA(x, a) = a$.

Next, we define a tree-based distance quantity between a node and a p -adic integer. For notational simplicity, we continue to use the symbol d for this quantity.

Definition 14. For a node $a \in \mathbb{T}_p$ and $x \in \mathbb{Z}_p$, the distance $d(x, a)$ between x and a is defined as

$$d(x, a) := 1/p^{\ell(LCA(x, a)) + 1}. \quad (11)$$

It follows from the above definitions that if a is an ancestor of x then $d(x, a) = 1/p^{\ell(a) + 1}$.

Denote by

$$\mathbb{Z}_{\geq 0} := \mathbb{N} \cup \{0\}$$

the set of non-negative integers. Define

$$J_p := \{1/p^k : k \in \mathbb{Z}_{\geq 0}\}$$

and let $d_a : \mathbb{Z}_p \rightarrow J_p$ be

$$d_a(x) := d(x, a). \quad (12)$$

2.3 Ball

For a p -adic integer x and a rational fraction $r = 1/p^k$, where $k \in \mathbb{Z}_{\geq 0}$, define a p -adic closed *ball* of radius r centered at x to be the set

$$B_r(x) := \{z : |z - x|_p \leq r\}. \quad (13)$$

Remark 15. By [3, Proposition 18.4], every point inside a ball may serve as its center. Every ball in $\mathbb{Z}_p$ is both closed and open.

We use an alternate description of a ball which is based on a node of $\mathbb{T}_p$ instead of a specific center point. For a node

$$a := [a_0, a_1, \dots, a_{k-1}] \in \mathcal{I}_p^k, \quad (14)$$

in $\mathbb{T}_p$, with level $\ell(a) = k - 1$, denote by

$$r_a = 1/p^{\ell(a) + 1}, \quad (15)$$

and let z be any p -adic integer whose expansion coefficients satisfy $z_i = a_i$, $0 \leq i \leq k - 1$. Denote by

$$B(a) := \{x : |x - z|_p \leq r_a\}. \quad (16)$$

This yields an alternative description of a ball because for any p -adic integer x and rational $r = 1/p^k$, a ball $B_r(x)$ can be represented as $B(a)$ by letting a be an ancestor node of x with $\ell(a) = k - 1$ and $r = r_a$. Henceforth, in (16), we choose the center z to be $(a_0, \dots, a_{k-1}, 0, \dots)$ which is represented by a leaf descendant of a (we choose this leaf for simplicity, however any of the infinitely many leaves that are descendants of a can be chosen).

Remark 16. For a node $a \in \mathbb{T}_p$, every $x \in B(a)$ is a descendant of a .

We denote by

$$\mathcal{B}_p := \{B(a) : a \in \mathbb{T}_p\} \quad (17)$$

the class of such balls.

For instance, if $a = [1, 2]$ then ball $B(a)$ is centered at p -adic $(1, 2, 0, \dots)$, its radius is $1/p^2$ and it contains any point $z \in \mathbb{Z}_p$ whose expansion satisfies $z_0 = 1$, $z_1 = 2$. If $a' = [1]$ then $B(a')$ is centered at $(1, 0, \dots)$, its radius $r = 1/p$ and it contains any point $z \in \mathbb{Z}_p$ whose $z_0 = 1$. These two balls are related as $B(a) \subset B(a')$.

Henceforth, we use a as in (14) for describing a ball $B(a)$ and $\alpha_k \in \mathcal{I}_{p^k}$ is defined as the corresponding sum

$$\alpha_k := \sum_{i=0}^{k-1} a_i p^i. \quad (18)$$

For a fixed k , each $a \in \mathcal{I}_p^k$ has a unique integer sum α_k hence a ball $B(a)$ can be identified by its unique corresponding integer α_k .

Definition 17. (*Descriptor*) For $a := [a_0, a_1, \dots, a_{k-1}] \in \mathcal{I}_p^k$ the *descriptor* of ball $B(a)$ is α_k .

The lower $\ell(a) = k - 1$, the smaller the range of possible values of α_k , and the larger the ball $B(a)$ whose descriptor is α_k .

Remark 18. From (18), the residue class corresponding to a ball $B(a)$ can be concisely expressed as $\alpha_k + p^k \mathbb{Z}_p$.

Definition 19. For a set $E \subset \mathbb{Z}_p$ and $x \in \mathbb{Z}_p$, define

$$d(x, E) := \inf_{z \in E} d(x, z).$$

Proposition 20. Let $a = [a_0, \dots, a_l] \in \mathbb{T}_p$ and $x \in \mathbb{Z}_p$ satisfy $x \notin B(a)$. Then

$$d(x, B(a)) = d(x, a).$$

Proof. Consider $x \notin B(a)$. Let $\tilde{a} := (a_0, \dots, a_l, 0, \dots)$, then $\tilde{a} \in B(a)$. For any $z \in B(a)$, we have $\ell(LCA(z, \tilde{a})) \geq \ell(a)$ because a is an ancestor of $\tilde{a}$ and z . Therefore

$$\begin{aligned} d(z, \tilde{a}) &= 1/p^{\ell(LCA(z, \tilde{a}))+1} && \text{(Proposition 11)} \\ &\leq 1/p^{\ell(a)+1} \\ &= r_a. && \text{(by (15))} \end{aligned}$$

Since every point of $B(a)$ may serve as its center (Remark 15), the assumption $x \notin B(a)$ implies that $d(x, z) > r_a$. Therefore $d(z, \tilde{a}) < d(x, z)$. By Proposition 2 it follows that $d(x, z) = d(x, \tilde{a})$ and therefore we have $\inf_{z \in B(a)} d(x, z) = d(x, \tilde{a})$. So

$$\begin{aligned} d(x, B(a)) &= \inf_{z \in B(a)} d(x, z) \\ &= d(x, \tilde{a}) \\ &= 1/p^{\ell(LCA(x, \tilde{a}))+1} && \text{(by Proposition 11)} \\ &= 1/p^{\ell(LCA(x, a))+1} \\ &= d(x, a) && \text{(by Definition 14).} \end{aligned}$$

□

3 p -adic concepts

In the Probably Approximately Correct (PAC) framework of learnability [30–32], a hypothesis is a binary classifier that can be used to infer some unknown concept from a finite sample.

By [3, Proposition 18.6] every open set in $\mathbb{Z}_p$ is a union of *disjoint* balls. This motivates the following definition of a p -adic concept.

Definition 21. (*p-adic concept*) A *p*-adic concept is a *finite* union U of balls $U = \bigcup_{a \in A} B(a)$, where A is a finite subset of distinct nodes of the tree $\mathbb{T}_p$.

Remark 22. While the above definition does not assume that balls $B(a)$ are disjoint, we note that if two balls in a representation of U are not disjoint, then U admits an equivalent representation as a union of fewer balls (by Proposition 3, two balls $B(a), B(a')$ that are not disjoint must have one contained within the other). Hence, given a union U , remove every $a \in A$ for which there exists $a' \in A$ with $B(a) \subseteq B(a')$. The remaining balls are maximal under inclusion (a ball $B(a)$ in the family $\{B(a) : a \in A\}$ is maximal if there is no $a' \in A$ such that $B(a) \subseteq B(a')$) thus the union is unchanged, and the remaining union of balls is automatically pairwise disjoint (because two distinct maximal balls cannot overlap in an ultrametric space).

Let us denote by $\mathcal{C}_n$ the class of *p*-adic concepts that consist of a union of up to n balls

$$\mathcal{C}_n := \left\{ U : U = \bigcup_{a \in A} B(a), |A| \leq n \right\}. \quad (19)$$

Next, we define binary classifiers on $\mathbb{Z}_p$ based on *p*-adic concepts.

From Section 3, a *p*-adic concept is a union $U \subseteq \mathbb{Z}_p$ of balls,

$$U := U_A = \bigcup_{a \in A} B(a). \quad (20)$$

Corresponding to U we define a binary *classifier* $h : \mathbb{Z}_p \rightarrow \{-1, 1\}$,

$$h_U(x) := \begin{cases} 1, & x \in U \\ -1, & \text{otherwise.} \end{cases} \quad (21)$$

A classifier h acts as an indicator function for the union of balls associated with a *p*-concept where instead of 0 we use -1 to denote not being in U .

For a finite integer $n \geq 1$ denote the family of binary classifiers based on *p*-adic concepts by

$$\mathcal{H}_n = \{h_U : U \in \mathcal{C}_n\}. \quad (22)$$

4 Width

For $U \subseteq \mathbb{Z}_p$, denote by $U^c := \mathbb{Z}_p \setminus U$ the complement of U . Next, we extend the notion of width introduced in [15] to functions $h \in \mathcal{H}_n$.

Definition 23. (*Width*) For $h := h_U \in \mathcal{H}_n$, denote by $w_h : \mathbb{Z}_p \rightarrow \{p^{-k} : k \in \mathbb{Z}_{\geq 0}\} \cup \{0\}$ the *width* $w_h(x)$ of h at x and define it as follows,

$$w_h(x) := \begin{cases} d(x, U^c), & x \in U \\ d(x, U), & x \in U^c. \end{cases}$$

The width of h at x is evaluated as follows: if $x \in U$, there exists a node a such that $x \in B(a)$. We assume throughout that the representation $U = \bigcup_{a \in A} B(a)$ is *reduced* in the sense that no ball $B(a)$ is strictly contained in another; this is without loss of generality by Remark 22. From Proposition 4 and Proposition 6, a ball in $\mathbb{Z}_p$ is open, and since U is a union of a finite number of balls, it is open, and hence its complement U^c is closed. Since $\mathbb{Z}_p$ is sequentially compact [27, Exercise 20, p.20] and is a

metric space, then it is compact [33, Theorem 28.2]. Since $\mathbb{Z}_p$ is compact and $U^{\mathfrak{C}} \subset \mathbb{Z}_p$ is closed, then $U^{\mathfrak{C}}$ is compact [34, Theorem 2.35] and the continuous function $s \mapsto d(x, s)$ attains its minimum on $U^{\mathfrak{C}}$ [34, Theorem 4.16]. Therefore there exists $z \in U^{\mathfrak{C}}$ such that $d(x, z) = d(x, U^{\mathfrak{C}})$. We claim that this z must be in some ball $B(a')$, where a' is a sibling of node a in $\mathbb{T}_p$ and a' is not in A .

To prove the claim, suppose no such sibling exists, then every sibling a' of a is in A and its ball is in U . This would imply that in addition to $B(a)$, the ball $B(\pi(a))$ is also in U which contradicts the reduced/minimal representation (Remark 22), namely, if all siblings are included, one may replace them by the parent ball and obtain a representation with fewer balls. Hence the claim is proved.

It follows that $LCA(x, z)$ is the parent node $\pi(a)$ of a . Therefore, we have

$$\begin{aligned} w_h(x) &= d(x, z) \\ &= \frac{1}{p^{\ell(LCA(x, z)) + 1}} \end{aligned} \tag{23}$$

$$\begin{aligned} &= \frac{1}{p^{\ell(\pi(a)) + 1}} \\ &= d(x, \pi(a)) \end{aligned} \tag{24}$$

$$= \frac{1}{p^{\ell(a)}} \tag{25}$$

$$= pr_a \tag{26}$$

where (23) follows from Proposition 11, (24) follows from Definition 14 since $\pi(a)$ is an ancestor of x , (25) follows from $\ell(\pi(a)) + 1 = \ell(a)$ and (26) follows from (15). If $x \notin U$ then $w_h(x)$ is the distance between x and its nearest p -adic integer z in U . Such z resides in a ball $B(a)$, for some $a \in A$. Therefore, the width of h at x is expressed as

$$\begin{aligned} w_h(x) &= d(x, U) \\ &= \inf_{z \in U} d(x, z) \quad (\text{by Definition 19}) \\ &= \min_{a \in A} \inf_{z \in B(a)} d(x, z) \\ &= \min_{a \in A} d(x, B(a)) \quad (\text{by Definition 19}) \\ &= \min_{a \in A} d(x, a) \quad (\text{by Proposition 20}) \\ &= \min_{a \in A} 1/p^{\ell(LCA(x, a)) + 1} \quad (\text{by Definition 14}) \\ &= 1/p^{\max_{a \in A} \ell(LCA(x, a)) + 1}. \end{aligned} \tag{27}$$

Hence, in summary, for a concept U_A , the width of classifier h at x is

$$w_h(x) = \begin{cases} pr_a & \text{if } h(x) = 1, x \in B(a) \text{ for some } a \in A, \\ 1/p^{\max_{a \in A} \ell(LCA(x, a)) + 1} & \text{if } h(x) = -1, x \notin \bigcup_{a \in A} B(a). \end{cases} \tag{28}$$

In case $h(x) = 1$ we can see from (28) that the *more shallow* (lower level ℓ) the LCA of x and a , where $B(a)$ is a ball that contains x or, in case $h(x) = -1$, the more shallow the deepest LCA of x and a over all balls $B(a)$ in the union, where no such ball contains x , the *larger* the width $w_h(x)$ of h at a point x .

In other words, if $h(x) = 1$, the more shallow the node a whose corresponding ball contains x , the lower $\ell(a)$, the larger the radius r_a , and the larger the width $w_h(x)$ at x . In case $h(x) = -1$, the more shallow the LCA of the closest node $a \in A$ to x , the lower its level, the larger its ball $B(a)$, and the larger the width $w_h(x)$ at x .

In both cases, a larger width is associated with a larger ball, and a smaller range of possible values of descriptor α by Definition 17. By (8), this means having a *simpler* residue class because α is limited to a smaller set of values.

This shows a relationship between the width of a classifier and a number-theoretic representation α of the ball of a concept based on which this width value is obtained, which is summarized by the next statement.

Remark 24. For p -adic concepts, width is proportional to a number-theoretic representation of simplicity.

In later sections, we show that there is an advantage in learning classifiers with a large width on a sample of points. If a learning algorithm obtains a classifier that has a large width on a sample of points, in effect, it obtains a simpler p -adic concept.

Denote by ξ a sample of labeled examples, $\xi := \{[x^{(i)}, y_i]\}_{i=1}^m$, where $x^{(i)} \in \mathbb{Z}_p$ and $y_i \in \{-1, 1\}$, $1 \leq i \leq m$.

In Definition 23 the width function $w_h(x)$ is the p -adic distance from x to the nearest oppositely classified region by h thus it takes a non-negative value.

When evaluating the width of an example $[x, y] \in \mathbb{Z}_p \times \{-1, 1\}$ it is useful to take into account not only x but also y , hence we define a signed version of the width function next.

Definition 25. (*Signed width*) Define the *signed width* $f_h(x)$ as

$$f_h(x) := h(x)w_h(x). \quad (29)$$

Next, we define the sample-width.

Definition 26. (*Sample width*) The sample width of h at ξ is defined as

$$w_\xi(h) := \min_{1 \leq i \leq m} y_i f_h(x^{(i)}).$$

If h misclassifies an example, then $h(x^{(i)}) \neq y_i$, and hence $y_i f_h(x^{(i)}) < 0$, which makes the sample width $w_\xi(h)$ negative. Otherwise, h correctly classifies the sample and the sample width is non-negative.

As mentioned above, in a subsequent section we show that learning large width classifiers yields lower error bounds hence next we give some examples of classifiers that have a maximum width.

Example 27. Consider a sample of two examples $\xi = \{[x_+, 1], [x_-, -1]\}$. We wish to find a classifier h based on a concept U that consists of a single ball such that the sample width $w_\xi(h)$ is large. Consider the $LCA(x_+, x_-)$, then it has child nodes a_+ and a_- such that $x_+ \in B(a_+)$, a_- is an ancestor of x_- and $x_- \notin B(a_+)$. Let us consider the classifier h whose concept is $B(a_+)$. By (28), the width at x_+ is $w_h(x_+) = 1/p^{\ell(a_+)} = 1/p^{\ell(LCA(x_+, x_-))+1}$ and $w_h(x_-) = 1/p^{\ell(LCA(x_-, a_+))+1} = 1/p^{\ell(LCA(x_+, x_-))+1}$. Any other ball $B(a)$ that contains x_+ with $\ell(a) < \ell(a_+)$ must contain x_- and hence a classifier based on $B(a)$ as a concept, will misclassify it as 1. Any other ball $B(a)$ that contains x_+ and has a larger level $\ell(a) > \ell(a_+)$, has a lower width $w_h(x_+)$ and the same value for $w_h(x_-) = 1/p^{\ell(LCA(x_+, x_-))+1}$ as the ball $B(a_+)$. Therefore the classifier h based on a concept with a single ball $B(a_+)$ yields the maximum sample width $w_\xi(h) = 1/p^{\ell(LCA(x_+, x_-))+1}$ over all possible h based on a single ball.

Example 28. Let $m_+, m_- \geq 1$ be finite integers. Consider a sample of m_+ positive examples $\xi_+ = \{[x_i^+, 1]\}_{i=1}^{m_+}$ examples, and a sample of m_- negative examples $\xi_- = \{[x_j^-, -1]\}_{j=1}^{m_-}$ and denote by $\xi = \xi_+ \cup \xi_-$. We wish to find a classifier h based on a concept U which has some finite number of balls, is consistent with the sample, namely, classifies it correctly, and has maximal sample width $w_\xi(h)$. For each positive example x_i^+ define

$$r_i := \max \{ \ell(LCA(x_i^+, x_j^-)) : 1 \leq j \leq m_- \}$$

and denote by $\hat{x}_i^-$ a negative example for which $\ell(LCA(x_i^+, \hat{x}_i^-)) = r_i$. Any concept that correctly classifies the sample cannot have width at x_i^+ larger than $1/p^{r_i+1}$ since otherwise the corresponding ball that contains x_i^+ would also contain $\hat{x}_i^-$. It follows that the sample width of any classifier which is consistent with the sample ξ is no larger than $\mu := \min \{1/p^{r_i+1} : 1 \leq i \leq m_+\}$.

Let us now present a procedure that finds a classifier with a large sample width. For each positive example x_i^+ find a negative example $\hat{x}_i^-$ with which it has the deepest LCA, as above; the same negative example may be associated with different positive examples. Select x_*^+ with its associated $\hat{x}^-$, denoted by x_*^- , that has the most shallow LCA,

$$\ell(LCA(x_*^+, x_*^-)) = \min \{ \ell(LCA(x_i^+, \hat{x}_i^-)) : 1 \leq i \leq m_+ \}.$$

Without loss of generality, suppose this example is $x_1^+ = x_*^+$ and its associated negative example is $x_1^- = x_*^-$. Now, follow the steps of Example 27 to construct a ball $B(a_1^+)$ based on a child node a_1^+ of $LCA(x_1^+, x_1^-)$ such that $x_1^+ \in B(a_1^+)$ and $x_1^- \notin B(a_1^+)$. Note that no other negative example is contained in $B(a_1^+)$ as implied by definition of $\hat{x}_1^-$, which means x_1^- has the deepest LCA with x_1^+ . Let $U = B(a_1^+)$. If all positive examples are contained in U then halt. Otherwise, from the remaining positive examples x_i^+ , $2 \leq i \leq m_+$, select x_*^+ with its associated $\hat{x}^-$, denoted by x_*^- , that has the most shallow LCA,

$$\ell(LCA(x_*^+, x_*^-)) = \min \{ \ell(LCA(x_i^+, \hat{x}_i^-)) : 2 \leq i \leq m_+ \}.$$

Without loss of generality, suppose this example is $x_2^+ = x_*^+$ and its associated negative example is $x_2^- = x_*^-$. As above, repeat the steps of Example 27 to construct a ball $B(a_2^+)$ and let the concept $U = U \cup B(a_2^+)$. And so on, repeat this process until U contains all positive examples. The number of balls in the union U is no larger than m_+ hence U consists of a finite number of balls. The resulting concept U satisfies:

- every ball constructed in the procedure excludes all negative examples
- every negative example is at distance at least μ from U
- every positive example is either selected at some stage and then gets width $1/p^{r_i+1}$ or is already covered by an earlier ball and then has width at least μ

hence the maximal sample width over all consistent classifiers representable by a finite union of balls is μ and the classifier h_U realizes this maximum.

5 Dimensions

In the current section we deal with a general distance space $\mathcal{X}$ and with a distance function $\rho : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$. We assume $\rho(x, x) = 0$ for all $x \in \mathcal{X}$, ρ is non-negative, and the space is bounded (for any $x, x' \in \mathcal{X}$, $\rho(x, x') \leq R$, where R is a real positive number). We do not assume that the distance function ρ satisfies the standard metric requirements of identity of indiscernible, symmetry, and triangle inequality [35].

We recall some notions in combinatorics of sets [25].

Definition 29. (*Touching*) Let $\mathcal{C} = \{E_j\}_{j=1}^m$ be a collection of sets $E_j \subset \mathcal{X}$. We say that $\mathcal{C}$ is *touched* by $\mathcal{X}$ if for every set A_i of the form

$$A_i = \bigcap_{j=1}^m E_j^*$$

there exists a point $z_i \in \mathcal{X}$ such that $z_i \in A_i$, $1 \leq i \leq 2^m$, where E_j^* is either E_j or its complement $E_j^c := \mathcal{X} \setminus E_j$.

Denote by $2^{\mathcal{X}}$ the family of all subsets of $\mathcal{X}$.

Definition 30. (*Independence number* [36]) Let $\mathcal{Q} \subset 2^{\mathcal{X}}$ be a (possibly infinite) family of sets in $\mathcal{X}$. The independence number (IN) of $\mathcal{Q}$, denoted by $IN(\mathcal{Q})$, is the cardinality of the *largest* collection $\mathcal{C} \subset \mathcal{Q}$ touched by $\mathcal{X}$. If this largest collection is not finite or if its cardinality is an arbitrarily large finite number then we define $IN(\mathcal{Q}) := \infty$.

A ball of radius r centered at a point x in $\mathcal{X}$ is defined as

$$B_r(x) := \{z : \rho(z, x) \leq r\}.$$

Denote the family of all balls $\mathcal{B} \subset 2^{\mathcal{X}}$ in $\mathcal{X}$ by

$$\mathcal{B} := \{B_r(x) : x \in \mathcal{X}, r > 0\}. \quad (30)$$

Definition 31. (*Combinatorial dimension*) The *combinatorial dimension* of a distance space $\mathcal{X}$ denoted by $\dim(\mathcal{X})$ is defined to be $IN(\mathcal{B})$.

Lemma 32. $\dim(\mathbb{Z}_p) = 1$.

Proof. From Proposition 6, $\mathbb{Z}_p$ is an ultrametric space hence Proposition 3 implies that, for any pair of balls B_1, B_2 in $\mathbb{Z}_p$ exactly one of the following holds: (i) $B_1 \cap B_2 = \emptyset$, (ii) $B_1 \subseteq B_2$, or (iii) $B_2 \subseteq B_1$. In case (i) the region $B_1 \cap B_2$ is empty. In case (ii), $B_1 \cap B_2^c = \emptyset$. In case (iii), $B_1^c \cap B_2 = \emptyset$. Hence $\{B_1, B_2\}$ is not touched. Therefore no pair of balls is touched and therefore $\dim(\mathbb{Z}_p) = IN(\mathcal{B}) = 1$. $\square$

Let $\mathbb{R}$ denote the set of real numbers. For $x \in \mathbb{R}$ denote the sign function by

$$\text{sgn}(x) := \begin{cases} -1 & \text{if } x \leq 0, \\ 1 & \text{if } x > 0. \end{cases}$$

For a real-valued function g , a set $S = \{x_i\}_{i=1}^m$ and a vector $r \in \mathbb{R}^m$, denote by

$$\text{sgn}(g|_S - r) := [\text{sgn}(g(x_1) - r_1), \dots, \text{sgn}(g(x_m) - r_m)].$$

Definition 33. (*Pseudo-dim*) For a class $\mathcal{G}$ of real-valued functions $g : \mathcal{X} \rightarrow \mathbb{R}$ the *Pseudo dimension* of $\mathcal{G}$ is defined as

$$Pdim(\mathcal{G}) := \max \{m \geq 1 : \exists r \in \mathbb{R}^m, \exists S \in \mathcal{X}^m, \{\text{sgn}(g|_S - r) : g \in \mathcal{G}\} = \{-1, 1\}^m\}. \quad (31)$$

For $g \in \mathcal{G}$ denote its max-norm on S by

$$\|g\|_S := \max_{x \in S} |g(x)|. \quad (32)$$

The ϵ -covering number of $\mathcal{G}$ with respect to the max-norm on S , denoted by $\mathcal{N}(\mathcal{G}, \epsilon, \|g\|_S)$, is defined as the *minimum* integer k such that there exists an ϵ -cover of $\mathcal{G}$ (with respect to this norm) of size k .

Define the uniform ϵ -covering number of $\mathcal{G}$ as

$$\mathcal{N}_\infty(\mathcal{G}, \epsilon, m) := \max_{S \subset \mathcal{X}: |S|=m} \mathcal{N}(\mathcal{G}, \epsilon, \|\cdot\|_S) \quad (33)$$

and if the maximum does not exist then we let it be infinite.

Definition 34. For $z \in \mathcal{X}$ define $d_z : \mathcal{X} \rightarrow [0, R]$ as

$$d_z(x) := \rho(z, x), \quad (34)$$

and define the class $\mathcal{D}$ of distance functions by

$$\mathcal{D} := \{d_z : z \in \mathcal{X}\}. \quad (35)$$

From [25, Theorem 4], the class $\mathcal{D}$ has

$$Pdim(\mathcal{D}) = dim(\mathcal{X}), \quad (36)$$

where the combinatorial dimension $dim(\mathcal{X})$ is defined in Definition 31.

From [11, Theorem 12.2], for $m \geq Pdim(\mathcal{D})$, the covering number of the class $\mathcal{D}$ is bounded using its Pseudo-dimension as follows,

$$\mathcal{N}_\infty(\mathcal{D}, \gamma, m) \leq \left(\frac{emR}{\gamma Pdim(\mathcal{D})} \right)^{Pdim(\mathcal{D})}. \quad (37)$$

Let

$$\mathcal{D}_p := \{d_a : a \in \mathbb{T}_p\}$$

denote a class of distance functions (12) on $\mathbb{Z}_p$. Under the identification $a \mapsto (a_0, \dots, a_{m-1}, 0, \dots) \in \mathbb{Z}_p$ (see paragraph following (16)), the class $\mathcal{D}_p$ is a subclass of $\{d_z : z \in \mathbb{Z}_p\}$ hence by Lemma 32 and (36) it follows that $Pdim(\mathcal{D}_p) \leq 1$. Let $a \in \mathbb{T}_p$ be an ancestor of x and $b \in \mathbb{T}_p$ be a node which is not an ancestor of x then $d(x, a) < d(x, b)$. Choose a threshold $r = d(x, a)$, then $\text{sgn}(d_a(x) - r) = -1$ and $\text{sgn}(d_b(x) - r) = 1$ hence $\{x\}$ is shattered by $\mathcal{D}_p$. With the previous inequality $Pdim(\mathcal{D}_p) \leq 1$ it follows that

$$Pdim(\mathcal{D}_p) = 1. \quad (38)$$

Substituting (38) for $Pdim(\mathcal{D})$ and (10) for R in (37) yields the bound

$$\mathcal{N}_\infty(\mathcal{D}_p, \gamma, m) \leq \frac{em}{\gamma}. \quad (39)$$

Definition 35. The class $\mathcal{D}_p$ is related to the class $\mathcal{B}_p$ of (17), or equivalently, the class $\mathcal{C}_1 = \mathcal{B}_p$ of single-ball p -adic concepts, the case $n = 1$ in Definition (19), because for a node $a = [a_0, \dots, a_l] \in \mathbb{T}_p$, we may describe a ball $B(a)$ in terms of a distance function (12) as follows,

$$B(a) = \{x : d_a(x) \leq r_a\}$$

where from (15), $r_a = 1/p^{\ell(a)+1}$. In the next section, we use this fact to define approximations of signed-width functions (Definition 25) of p -adic concepts.

6 $\mathcal{F}_n$

In this section we construct finite approximating families for the class $\mathcal{F}_n$ of signed-width functions associated with p -adic concepts consisting of unions of at most n balls. The purpose of this construction is to obtain explicit covering-number bounds for $\mathcal{F}_n$, which are needed in the next section to derive generalization bounds for large-width classification on $\mathbb{Z}_p$. The construction combines two ingredients developed earlier: the low complexity of the distance-function class $\mathcal{D}_p$, whose pseudo-dimension equals 1, and the hierarchical structure induced by p -adic balls on a finite sample. In particular, intersections

of balls with a sample generate a dendrogram-like partition structure, and counting these admissible partitions yields a finite cover of $\mathcal{F}_n$ whose cardinality can be bounded explicitly.

From (24), (27) and (28), the signed width function (29) corresponding to a concept (20) is expressed in terms of functions d_a in $\mathcal{D}_p$, as follows

$$f_h(x) := f_A(x) = \begin{cases} d_{\pi(a)}(x) & \text{if } x \in B(a) \text{ for some } a \in A, \\ -\min_{a \in A} d_a(x) & \text{if } x \notin \bigcup_{a \in A} B(a), \end{cases} \quad (40)$$

where $A \subset \mathbb{T}_p$ has cardinality $|A| \leq n$, and is a set of nodes a that correspond to balls $B(a)$ in the union $\bigcup_{a \in A} B(a)$ that defines a concept U in the class $\mathcal{C}_n$ of p -adic concepts.

The class of signed width functions (29) is denoted by

$$\mathcal{F}_n = \{f_h : h \in \mathcal{H}_n\} \quad (41)$$

where $\mathcal{H}_n$ is defined in (22), or alternatively, $\mathcal{F}_n$ is a class of all functions f_A where $A \subset \mathbb{T}_p$, $|A| \leq n$ and the balls $B(a)$, $a \in A$, are disjoint.

Fix a set $S = \{x_1, \dots, x_m\} \subset \mathbb{Z}_p$. Let

$$\hat{D}_\epsilon := \{\hat{d}_1, \dots, \hat{d}_{M_\epsilon}\}$$

be an ϵ -cover of $\mathcal{D}_p$ of minimal cardinality

$$M_\epsilon := \left| \hat{D}_\epsilon \right| \quad (42)$$

with respect to the norm (32).

From (33), we have

$$M_\epsilon = \mathcal{N}(\mathcal{D}_p, \epsilon, \|\cdot\|_S) \leq \mathcal{N}_\infty(\mathcal{D}_p, \epsilon, m). \quad (43)$$

Let

$$E = \{e_1, \dots, e_n\} \subset \mathbb{T}_p, \quad k := [k_1, \dots, k_n], \quad l := [l_1, \dots, l_n] \quad (44)$$

with $k_i, l_i \in \{1, 2, \dots, M_\epsilon\}$, $1 \leq i \leq n$, such that the balls $B(e_i)$, $1 \leq i \leq n$, are disjoint. We define $\hat{f} : S \rightarrow J_p$, where for $x \in S$,

$$\hat{f}_{E,k,l}(x) := \begin{cases} \hat{d}_{k_i}(x) & \text{if } x \in B(e_i) \cap S \text{ for some } e_i \in E \\ -\min_{1 \leq i \leq n} \hat{d}_{l_i}(x) & \text{if } x \in S \setminus \bigcup_{1 \leq i \leq n} B(e_i). \end{cases} \quad (45)$$

Next, we use (45) to construct an approximation for f_A over S .

Proposition 36. *Let $S = \{x_1, x_2, \dots, x_m\} \subset \mathbb{Z}_p$ and $A = \{a_1, \dots, a_n\} \subset \mathbb{T}_p$. Let $f_A \in \mathcal{F}_n$. Define a selector $\sigma : \mathcal{D}_p \rightarrow \{1, 2, \dots, M_\epsilon\}$ where for $a_i \in A$,*

$$j_i := \sigma(d_{a_i}), \quad j_i^\pi := \sigma(d_{\pi(a_i)}) \quad (46)$$

and

$$\left\| d_{a_i} - \hat{d}_{j_i} \right\|_S \leq \epsilon, \quad \left\| d_{\pi(a_i)} - \hat{d}_{j_i^\pi} \right\|_S \leq \epsilon,$$

where $\hat{d}_{j_i}, \hat{d}_{j_i^\pi} \in \hat{D}_\epsilon$, $1 \leq i \leq n$. In (45), substitute

$$E = A, \quad k = j^\pi, \quad l = j \quad (47)$$

to obtain $\hat{f}_{A,j,j^\pi} : S \rightarrow J_p$. Then

$$\left\| f_A - \hat{f}_{A,j,j^\pi} \right\|_S \leq \epsilon.$$

Proof. Fix $x \in S$ and consider the case that $x \in B(a_i)$, for some $a_i \in A$. Then

$$\begin{aligned}
\left| f_A(x) - \hat{f}_{A,j,j^\pi}(x) \right| &= \left| d_{\pi(a_i)}(x) - \hat{d}_{\sigma(d_{\pi(a_i)})}(x) \right| \\
&= \left| d_{\pi(a_i)}(x) - \hat{d}_{j_i^\pi}(x) \right| \\
&\leq \left\| d_{\pi(a_i)} - \hat{d}_{j_i^\pi} \right\|_S \\
&\leq \epsilon.
\end{aligned} \tag{48}$$

Now, consider the case $x \in S \setminus \bigcup_{a \in A} B(a)$. Then

$$\begin{aligned}
\left| f_A(x) - \hat{f}_{A,j,j^\pi}(x) \right| &= \left| -\min_{1 \leq i \leq n} d_{a_i}(x) - \left(-\min_{1 \leq i \leq n} \hat{d}_{\sigma(d_{a_i})}(x) \right) \right| \\
&= \left| \min_{1 \leq i \leq n} \hat{d}_{j_i}(x) - \min_{1 \leq i \leq n} d_{a_i}(x) \right|.
\end{aligned} \tag{49}$$

Denote by $l := \operatorname{argmin}_{1 \leq i \leq n} d_{a_i}(x)$. Then

$$\begin{aligned}
\min_{1 \leq i \leq n} \hat{d}_{j_i}(x) - \min_{1 \leq i \leq n} d_{a_i}(x) &= \min_{1 \leq i \leq n} \hat{d}_{j_i}(x) - d_{a_l}(x) \\
&\leq \hat{d}_{j_l}(x) - d_{a_l}(x) \\
&\leq \max_{1 \leq i \leq n} \left| \hat{d}_{j_i}(x) - d_{a_i}(x) \right|.
\end{aligned} \tag{50}$$

Similarly, let $l := \operatorname{argmin}_{1 \leq i \leq n} \hat{d}_{j_i}(x)$. Then

$$\begin{aligned}
\min_{1 \leq i \leq n} d_{a_i}(x) - \min_{1 \leq i \leq n} \hat{d}_{j_i}(x) &= \min_{1 \leq i \leq n} d_{a_i}(x) - \hat{d}_{j_l}(x) \\
&\leq d_{a_l}(x) - \hat{d}_{j_l}(x) \\
&\leq \max_{1 \leq i \leq n} \left| d_{a_i}(x) - \hat{d}_{j_i}(x) \right|.
\end{aligned} \tag{51}$$

For any $x \in S$ we have $\left| d_{a_i}(x) - \hat{d}_{j_i}(x) \right| \leq \left\| d_{a_i} - \hat{d}_{j_i} \right\|_S$ and since $\hat{D}_\epsilon$ covers $\mathcal{D}_p$ then for any $i \in \{1, \dots, n\}$, $\left\| d_{a_i} - \hat{d}_{j_i} \right\|_S \leq \epsilon$. It follows that (49) and (50) are bounded from above by ϵ and so is (51). With (48), it follows that for any $x \in S$, $\left| f_A(x) - \hat{f}_{A,j,j^\pi}(x) \right| \leq \epsilon$ therefore $\left\| f_A - \hat{f}_{A,j,j^\pi} \right\|_S \leq \epsilon$. $\square$

Two-step construction of $\hat{f}$: Fix a sample $S = \{x_1, \dots, x_m\} \subset \mathbb{Z}_p$. In constructing $\hat{f} : S \rightarrow J_p$ according to (46) and (47), the value of $\hat{f}$ on S can be obtained through two steps:

- define a partition $\Pi_L(S) = \{R_j\}_{j=0}^L$ of S into disjoint sets $R_j \subseteq S$ of the form

$$R_j = S \cap B(e_i) \neq \emptyset \tag{52}$$

for some $e_i \in E$, $1 \leq j \leq L \leq n$, and

$$R_0 = S \setminus \bigcup_{1 \leq i \leq n} B(e_i) \tag{53}$$

which may be empty. (L may be smaller than n because of Remark 22.)

- choose a value in J_p for $\hat{f}$ on each non-empty region R_j , $0 \leq j \leq L$.

According to (45), for each $x \in S$, the value of $\hat{f}(x)$ depends on whether x is in R_0 or in R_j , $1 \leq j \leq L$, and is determined by a function in the minimal ϵ -cover $\hat{D}_\epsilon$ of $\mathcal{D}_p$, whose index belongs to the set

$$K := \{1, 2, \dots, M_\epsilon\} \quad (54)$$

where M_ϵ is defined in (42). Hence, a function $\hat{f}_E$ corresponds to a partition and a combination of values from the set J_p assigned to the parts of this partition. Since there are a finite number of partitions $\Pi_L(S)$ of S , and a finite number of choices for values that $\hat{f}$ may take on each part, there are a finite number of mappings $\hat{f}_{E,k,l}$ defined in (45), despite the fact that there are infinitely many subsets $E \subset \mathbb{T}_p$.

Before stating and proving the next result we define some notions that we need for the proof.

For two subsets C, C' of S define the distance

$$\text{dist}(C, C') := \min_{x \in C, x' \in C'} d(x, x')$$

where d is defined in (9). Define

$$\text{diam}(C) := \max_{x \in C, x' \in C} d(x, x').$$

Let

$$r_1 < r_2 < \dots < r_v,$$

where $1 \leq v \leq \binom{m}{2}$, be all possible distinct distances between pairs of elements x_i, x_j of S and let $r_0 := 0$.

We construct a tree $\mathcal{T}_S$, called *dendrogram*, whose nodes are subsets of S and are referred to as clusters. We define $\mathcal{T}_S$ recursively: the base level $\mathcal{C}^{(0)} := \{\{x_j\} : 1 \leq j \leq m\}$ consists of clusters that are leaves of $\mathcal{T}_S$. At level $i \in \{1, \dots, v\}$, nodes of $\mathcal{T}_S$ are clusters $C \in \mathcal{C}^{(i)}$. A subset $C \subseteq S$ is a *cluster* at level $\mathcal{C}^{(i)}$ if there exists a sub-collection of clusters from the previous level, $\mathcal{Q} \subseteq \mathcal{C}^{(i-1)}$, satisfying the following conditions:

- $C = \bigcup_{Q \in \mathcal{Q}} Q$
- for all $Q_a, Q_b \in \mathcal{Q}$, $\text{dist}(Q_a, Q_b) \leq r_i$
- there exists a pair $Q_a, Q_b \in \mathcal{Q}$, $\text{dist}(Q_a, Q_b) = r_i$, which implies that $\text{diam}(C) = r_i$. (This condition ensures that cluster C is formed specifically at level i due to reaching a threshold value of r_i , rather than being a pre-existing cluster from a lower level that is carried over.)

A point $x_i \in S$ belongs to multiple clusters at different levels. At level i , a cluster $C \in \mathcal{C}^{(i)}$ may be a pre-existing cluster $C' \in \mathcal{C}^{(i-1)}$ of level $i-1$ if the distance from C' to any other cluster in $\mathcal{C}^{(i-1)}$ is greater than r_i .

We say that two clusters are *disjoint* at level i if there is no cluster $C \in \mathcal{C}^{(i)}$ that contains both of them.

Lemma 37. *Let $S = \{x_1, \dots, x_m\} \subset \mathbb{Z}_p$ and $E = \{e_1, \dots, e_n\} \subset \mathbb{T}_p$. The number of functions $\hat{f}_{E,k,l} : S \rightarrow J_p$ is no more than*

$$M_\epsilon^n \sum_{L=0}^n \binom{2m-1}{L} M_\epsilon^L \quad (55)$$

where M_ϵ is defined in (42).

Proof. Since a function $\hat{f}$ corresponds to a partition of S and a combination of values over its parts, to bound from above the number of functions $\hat{f}$ we start by obtaining an upper bound on the number of partitions $\Pi_L(S) = \{R_0, \dots, R_L\}$ of S into $L \leq n$ parts that are picked by functions $\hat{f}$, where R_i , $1 \leq i \leq L$, are of the form (52) and the part $R_0 = S \setminus \bigcup_{1 \leq i \leq n} B(e_i)$ may be empty. If $L \geq 1$, the parts $R_i \subseteq S$, $1 \leq i \leq L$, are non-empty sets of the form $S \cap B(e_i)$, $e_i \in E$, $1 \leq i \leq L$. If $L = 0$, the partition consists of a single part $R_0 = S$.

To bound the number of partitions $\Pi_L(S)$ it suffices to find the maximal number of combinations of subsets of S picked out by unions of $L \leq n$ balls $B \in \mathcal{B}_p$; if there are elements of S that are not picked, they form part R_0 . Each such combination of subsets of S is a distinct partition $\Pi_L(S)$.

Next, we show that single balls $B \in \mathcal{B}_p$ pick out subsets of S that are clusters C .

Claim 38. Let $B_r \in \mathcal{B}_p$ be a ball of radius r . If $B_r \cap S$ is not empty, then $B_r \cap S = C$ where $C \in \mathcal{C}^{(i)}$, $i = \operatorname{argmax} \{j : 0 \leq j \leq v, r_j \leq r\}$ (that is, $0 = r_0 < r_1 < r_2 < \dots < r_i \leq r < r_{i+1} < \dots < r_v$).

Proof of the claim: The proof is divided into two parts. *Part (i):* We show that if a point $x \in S$ belongs to B_r then B_r contains the whole cluster of maximum-level i , $r_i \leq r$, that contains x . Consider $x \in B_r \cap S$. From above, x belongs to multiple clusters at different levels. Let $i \in \{0, 1, \dots, v\}$ be the maximum level such that $r_i \leq r$ and such that there exists a cluster $C \in \mathcal{C}^{(i)}$ that contains x . If there is no $x' \neq x$ such that $x' \in C$ then $B_r \cap S = C = \{x\}$. If there is another point $x' \in C$ then consider a center point $\tilde{x} \in B_r$ which is not necessarily in S (by Remark 15, every point in B_r is a center). Then $d(\tilde{x}, x') \leq \max\{d(\tilde{x}, x), d(x, x')\} \leq \max\{r, r_i\} = r$, which follows from x and x' being in the same cluster C hence their distance is no more than r_i , and from the fact that $x, \tilde{x}$ are in the same ball B_r therefore their distance is no more than r (13). It follows that $d(\tilde{x}, x') \leq r$ and therefore $x' \in B_r$. Hence if B_r picks out a point $x \in S$, then it picks out every other point in its maximum-level cluster $C \in \mathcal{C}^{(i)}$. Therefore $C \subseteq B_r \cap S$.

Part (ii): Next, we show that $B_r \cap S$ does not contain disjoint clusters. Suppose on the contrary that there are two disjoint clusters C and C' contained in $B_r \cap S$. Since $r_i \leq r$, C and C' are disjoint as clusters in $\mathcal{C}^{(i)}$ hence there is no cluster at level i that contains C and C' . This implies that $\operatorname{dist}(C, C') \geq r_{i+1}$. This implies that there exist $x \in C$ and $x' \in C'$ such that $d(x, x') \geq r_{i+1} > r$. Hence $d(x, x') \not\leq r$ which implies that either x or x' is not contained in $B_r \cap S$. This contradicts the assumption that C and C' are subsets of $B_r \cap S$. Therefore there are no disjoint clusters in $B_r \cap S$.

From both parts above, it follows that $B_r \cap S = C$ where C is a single cluster $C \in \mathcal{C}^{(i)}$, as claimed. $\square$

It follows from Claim 38 that subsets of S picked by the family $\mathcal{B}_p$ of p -adic balls are single whole clusters $C \subseteq S$, and by definition, such clusters correspond to nodes of $\mathcal{T}_S$. Hence the number of subsets of S picked by the family $\mathcal{B}_p$ equals the number of nodes of $\mathcal{T}_S$.

Next, we show that the number of non-leaf nodes of $\mathcal{T}_S$ is no larger than $m - 1$.

To bound the total number of nodes of the dendrogram $\mathcal{T}_S$ associated with S , it is enough to consider a set S' of the same cardinality m whose dendrogram has the largest possible number of non-leaf nodes. Indeed, any dendrogram built from m points has at most as many non-leaf nodes as such an extremal dendrogram. We may therefore choose S' so that its dendrogram is maximal, namely, each new merge appends a singleton to a previously formed cluster. The set S' has maximal collection of clusters: the first cluster C_1 has a minimal cardinality of 2 and corresponds to a node q_1 at level 1 which is a parent of two leaves $\{x_1\}$ and $\{x_2\}$, thus $C_1 = \{x_1, x_2\}$. The next node q_2 is at level 2 and is a parent of q_1 and a third leaf $\{x_3\}$. It corresponds to a cluster $C_2 = C_1 \cup \{x_3\} = \{x_1, x_2, x_3\}$. This is repeated until we get to level $m - 1$, where node q_{m-1} is a parent of q_{m-2} and leaf $\{x_m\}$ and corresponds to a cluster $C_{m-1} = C_{m-2} \cup \{x_m\} = S$. The dendrogram $\mathcal{T}_{S'}$ has exactly m leaves and $m - 1$ non-leaf nodes, hence $2m - 1$ nodes in total. This bounds from above the number of nodes in the dendrogram $\mathcal{T}_S$ associated with S and hence the number of subsets of S picked by the family $\mathcal{B}_p$. Therefore the total number of non-empty clusters of *any* cardinality obtained by intersecting balls $B \in \mathcal{B}_p$ with S is no more than $2m - 1$.

It follows that a partition $\Pi_L(S)$ consists of L subsets of S , referred to as parts R_i , $1 \leq i \leq L \leq n$, that are clusters picked by n p -adic balls and a part R_0 which consists of the remaining elements of S . A partition therefore corresponds to a combination of $L \leq n$ clusters. We have $L \leq n$ and the number of clusters is at most $2m - 1$, hence there are no more than $\sum_{k=0}^n \binom{2m-1}{k}$ possible combinations of $L \leq n$ clusters. The case $k = 0$ represents a partition $\Pi_L(S)$ with a single part $R_0 = S$ which is not picked by any of the n balls.

Next, we count the number of functions that can be constructed on a partition of S . Fix a partition $\Pi_L(S)$ of k parts. On each non-empty part R_i of $\Pi_L(S)$, we define the values for $\hat{f}$ as follows: if R_0 is not empty then according to Proposition 36, for $x \in R_0$, we let $\hat{f}(x) := -\min_{1 \leq i \leq n} \hat{d}_{j_i}(x)$ where $j := [j_1, j_2, \dots, j_n] \in K^n$, and K is defined in (54). Thus it suffices to select a value for vector j in order to specify the value of $\hat{f}(x)$ over R_0 (this choice of j holds uniformly for all x in R_0). There are no more than M_ϵ^n possible values for vector j . If $x \in R_i$, $1 \leq i \leq L$, by Proposition 36, $\hat{f}(x) = \hat{d}_{j_i}(x)$, so for each R_i a single function from K is chosen. Hence there are no more than M_ϵ^L possible combinations of values that a function $\hat{f}$ may take over these L parts.

Thus for a fixed value of L and a fixed partition $\Pi_L(S) = \{R_0, \dots, R_L\}$, there are no more than M_ϵ^{n+L} possible functions $\hat{f}$ defined on $\Pi_L(S)$. For $L \leq n$, the number of functions on any partition $\Pi_L(S)$ is bounded from above by

$$M_\epsilon^n \sum_{L=0}^n \binom{2m-1}{L} M_\epsilon^L.$$

□

Given a sample $S \subset \mathbb{Z}_p$ and $f := f_A \in \mathcal{F}_n$, the two-step construction on page 16 gives an ϵ -approximation $\hat{f}_{A,j,j^\pi}$ of f ,

$$\|f_A - \hat{f}_{A,j,j^\pi}\|_S \leq \epsilon. \quad (56)$$

Lemma 37 implies that the number of functions $\hat{f}_{A,j,j^\pi}$ is bounded from above by (55). Hence, with respect to the max-norm on S , there exists a finite ϵ -cover of $\mathcal{F}_n$, denoted by

$$\hat{F}_\epsilon := \{\hat{f}_1, \hat{f}_2, \dots, \hat{f}_{N_\epsilon}\}. \quad (57)$$

Lemma 39. (Covering number of $\mathcal{F}_n$) Let $S \subset \mathbb{Z}_p$ be of cardinality $|S| = m$. Let M_ϵ be the covering number of $\mathcal{D}_p$ with respect to the max-norm on S (32). If $m \geq \lceil (n+1)/2 \rceil$ then the covering number $\mathcal{N}(\mathcal{F}_n, \epsilon, \|\cdot\|_S)$ of $\mathcal{F}_n$ with respect to the max-norm on S is no larger than

$$\mathcal{N}(\mathcal{F}_n, \epsilon, \|\cdot\|_S) \leq N_\epsilon := \left(\frac{e M_\epsilon^2 (2m-1)}{n} \right)^n. \quad (58)$$

Proof. Fix a set $S \subset \mathbb{Z}_p$ of cardinality m . From Proposition 36, for any $f \in \mathcal{F}_n$ there exists an ϵ -approximation $\hat{f} \in \hat{F}_\epsilon$ of f with respect to the $\|\cdot\|_S$ norm, where $\hat{F}_\epsilon$ is defined in (57). Therefore $\hat{F}_\epsilon$ is an ϵ -cover of $\mathcal{F}_n$ with respect to this norm and its cardinality is bounded from above by

$$\begin{aligned} M_\epsilon^n \sum_{k=0}^n \binom{2m-1}{k} M_\epsilon^k &\leq M_\epsilon^{2n} \sum_{k=0}^n \binom{2m-1}{k} \\ &\leq M_\epsilon^{2n} \left(\frac{e(2m-1)}{n} \right)^n \\ &= \left(\frac{e M_\epsilon^2 (2m-1)}{n} \right)^n \end{aligned} \quad (59)$$

where (59) holds if $m \geq \lceil (n+1)/2 \rceil$. □

7 Learning

Let P be any probability measure on some appropriate σ -algebra over the set $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} := \{-1, 1\}$. Functions on $\mathcal{Z}$ are assumed to be measurable without explicit reference. A sample is a finite set of labeled points $\{z_i\}_{i=1}^m$, $z_i = (x_i, y_i) \in \mathcal{X} \times \mathcal{Y}$, $1 \leq i \leq m$. The space of all samples of size m is denoted by $\mathcal{Z}^m$. A *random* sample $\xi := \{(X_i, Y_i)\}_{i=1}^m \subset \mathcal{Z}$ is drawn according to the joint (product) probability measure P^m which is defined on $\mathcal{Z}^m$, that is, ξ consists of m i.i.d. random pairs (X_i, Y_i) , $1 \leq i \leq m$, each drawn according to P .

We consider the problem of learning binary classification using the class $\mathcal{H}_n$ of classifiers based on p -adic concepts on $\mathbb{Z}_p$. By learning we mean the process of finding a good classifier h_U based on a p -adic concept U which has a low probability of misclassifying a randomly drawn labeled point $(X, Y) \in \mathcal{Z}$. This misclassification error is referred to as the *generalization error* of h_U and is defined as the following probability

$$P(h_U(X) \neq Y). \quad (60)$$

Our aim in this section is to obtain an upper bound on the generalization error of any h_U in $\mathcal{H}_n$.

A binary classifier $h := h_U$ can be expressed in terms of a signed width function (40) as follows

$$h(x) = \text{sgn}(f(x)) \quad (61)$$

where $f := f_A$, and A is the set of nodes in the definition (20) of concept U .

We choose to represent the classifier in terms of a real-valued function as in (61) because the error can then be bounded by a quantity which involves the covering number of the class $\mathcal{F}_n$ (41) which is bounded from above by an expression that depends on the Pseudo-dimension of $\mathcal{D}_p$. While a more direct approach to the analysis of learning $\mathcal{H}_n$ would involve the VC-dimension of $\mathcal{H}_n$ instead of the Pseudo-dimension of $\mathcal{F}_n$, this would leave us with an error bound that does not depend on the width parameter γ and such bounds are less tight.

We can express the generalization error of h in terms of f_A as follows,

$$\begin{aligned} P(h(X) \neq Y) &= P(Yf(X) < 0) + P(Y = 1, f(X) = 0) \\ &\leq P(Yf(X) \leq 0). \end{aligned} \quad (62)$$

Thus learning $\mathcal{H}_n$ amounts to learning the class $\mathcal{F}_n$, that is, the aim is to find an f for which (62) is low and this ensures that the error of h is low.

Formally, under the extension of the PAC model of learnability to real-valued functions [37] the learning problem is as follows: let $\mathcal{Z} := \mathbb{Z}_p \times \{-1, 1\}$. We wish to learn a binary classifier h by finding an f from the class $\mathcal{F}_n$ using a randomly drawn training sample $\xi := \{(X_i, Y_i)\}_{i=1}^m \subset \mathcal{Z}$, such that f has a small probability value (62) and therefore its corresponding h has a small error. This value (62) cannot be measured because the probability distribution P is assumed to be unknown so we need an empirical estimate. We follow the notion of large-margin learning [11, Ch. 13] and for $\gamma \in (0, \text{diam}(\mathbb{Z}_p)]$ we define the following empirical quantity,

$$P_m(Yf(X) \leq \gamma) := \frac{1}{m} \sum_{i=1}^m \mathbb{I}\{Y_i f(X_i) \leq \gamma\} \quad (63)$$

which we refer to as the *empirical γ -margin error* of f . The next result shows that given any γ in the range of possible margin values, with high confidence, uniformly over all $h \in \mathcal{H}_n$ the generalization error of h is not much greater than the empirical γ -margin error of its corresponding f .

Lemma 40. Let $\mathcal{H}_n$ be a class of binary classifiers $h(x) = \text{sgn}(f(x))$ where

$$f(x) := f_A(x) = \begin{cases} d_{\pi(a)}(x) & \text{if } x \in B(a) \text{ for some } a \in A, \\ -\min_{a \in A} d_a(x) & \text{if } x \notin \bigcup_{a \in A} B(a), \end{cases}$$

corresponds to a p -adic concept $U_A = \bigcup_{a \in A} B(a)$ with $A \subset \mathbb{T}_p$, $|A| \leq n$. Denote the class of all such functions f by $\mathcal{F}_n$. Let P be any probability measure on $\mathcal{Z} := \mathbb{Z}_p \times \{-1, 1\}$ and let $\delta \in (0, 1]$ be a confidence parameter. For any integer $m \geq \lceil (n+1)/2 \rceil$ and $\gamma \in (0, 1]$ the following holds with P^m -probability at least $1 - \delta$ for a random training sample $\xi \in \mathcal{Z}^m$ drawn i.i.d. according to P^m : for all $f \in \mathcal{F}_n$,

$$P(h(X) \neq Y) \leq P_m(Yf(X) \leq \gamma) + \sqrt{\frac{8}{m} \left(n \ln \left(\frac{16e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{2}{\delta} \right) \right)}. \quad (64)$$

Proof. Define a margin function

$$\mu(f(x), y) := yf(x). \quad (65)$$

In [11], Theorem 10.1 is stated for a class of real valued functions that map a domain into the unit interval, with a different margin function (Definition 9.1, [11]) and binary classification values $\mathcal{Y} = \{0, 1\}$. The proof of that theorem holds for our setup by substituting $\mathcal{F}_n$, the margin function (65) and binary values $y \in \{-1, 1\}$, for the corresponding three quantities in their setup (the definition of empirical γ -margin error is the same in both setups). Therefore, the theorem yields the following bound,

$$P^m(\exists f \in \mathcal{F}_n, P(Yf(X) \leq 0) > P_m(Yf(X) \leq \gamma) + \epsilon) \leq 2\mathcal{N}_\infty(\mathcal{F}_n, \gamma/2, 2m) \exp(-\epsilon^2 m/8). \quad (66)$$

From Lemma 39, the right hand of (58) holds for any $S \subset \mathbb{Z}_p$ of cardinality m , hence it follows that the uniform covering number (33) is bounded as

$$\begin{aligned} \mathcal{N}_\infty(\mathcal{F}_n, \gamma/2, 2m) &\leq M_{\gamma/2}^{2n} \left(\frac{e(4m-1)}{n} \right)^n \\ &\leq \left(\frac{4em}{\gamma} \right)^{2n} \left(\frac{e(4m-1)}{n} \right)^n \end{aligned} \quad (67)$$

where (67) follows from (39) and (43).

This implies that the right hand side of (66) is bounded from above by the expression

$$2 \left(\frac{4em}{\gamma} \right)^{2n} \left(\frac{e(4m-1)}{n} \right)^n \exp(-\epsilon^2 m/8)$$

which is simplified to

$$2 \left(\frac{16e^3 m^2 (4m-1)}{\gamma^2 n} \right)^n \exp(-\epsilon^2 m/8).$$

Setting this to be δ and solving for ϵ yields (64). $\square$

Next is the main result of this section, which states that with a large confidence, uniformly over all possible values of γ and all classifiers $h \in \mathcal{H}_n$, the generalization error of h is not much greater than the empirical γ -margin error of its corresponding f .

Theorem 41. *With the same setup as in Lemma 40, let P be any probability measure on $\mathbb{Z}_p \times \{-1, 1\}$ and let $\delta \in (0, 1]$ be a confidence parameter. For any integer $m \geq \lceil (n+1)/2 \rceil$ the following holds with P^m -probability at least $1 - \delta$ for a random training sample $\xi \in \mathcal{Z}^m$ drawn i.i.d. according to P^m : for all $f \in \mathcal{F}_n$ and all $\gamma \in (0, 1]$,*

$$P\left(h(X) \neq Y\right) \leq P_m\left(Yf(X) \leq \gamma\right) + \sqrt{\frac{8}{m} \left(n \ln \left(\frac{64e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{4}{\gamma \delta} \right) \right)}. \quad (68)$$

Note that in Theorem 41, γ is not specified in advance of drawing the sample ξ . In particular, choosing γ to be the largest value for which the empirical γ -margin error on ξ is 0 implies that h is consistent with the sample (classifies it correctly), and that γ is precisely the sample width of h on ξ .

Proof. We consider the bad event that (62) is *not* less than or equal to the sum of (63) plus some small deviation ϵ , that is, the event that

$$P\left(Yf(X) \leq 0\right) > P_m\left(Yf(X) \leq \gamma\right) + \epsilon.$$

This can be rephrased as the following event,

$$P\left(Yf(X) > 0\right) < P_m\left(Yf(X) > \gamma\right) - \epsilon. \quad (69)$$

Now, consider the bad event that there exists some $f \in \mathcal{F}_n$ and some value of $\gamma \in (0, 1]$ such that (69) occurs with $\epsilon := \epsilon(\gamma/2, \gamma\delta/2)$, where $\epsilon(a, b)$ is decreasing with a and b and is specified further below. Define the set $E(\gamma, \delta) \subset \mathcal{Z}^m$,

$$E(\gamma, \delta) := \{\xi : \exists f \in \mathcal{F}_n, P(Yf(X) > 0) < P_m(Yf(X) > \gamma) - \epsilon(\gamma/2, \gamma\delta/2)\}. \quad (70)$$

The probability of the bad event therefore equals $P^m\left(\bigcup_{\gamma \in (0, 1]} E(\gamma, \delta)\right)$.

We follow a technique of [38] (see [11, Lemma 15.5]). Let $\gamma_0 := \text{diam}(\mathbb{Z}_p) = 1$ and for $l = 1, 2, \dots, \infty$, define

$$\gamma_l := \left(\frac{1}{2}\right)^l$$

and

$$\Gamma_l := (\gamma_l, \gamma_{l-1}].$$

We have

$$\begin{aligned} P^m\left(\bigcup_{\gamma \in (0, 1]} E(\gamma, \delta)\right) &= P^m\left(\bigcup_{l=1}^{\infty} \bigcup_{\gamma \in \Gamma_l} E(\gamma, \delta)\right) \\ &\leq \sum_{l=1}^{\infty} P^m\left(\bigcup_{\gamma \in \Gamma_l} E(\gamma, \delta)\right). \end{aligned} \quad (71)$$

Define the set

$$E_l := \left\{ \xi : \exists f \in \mathcal{F}_n, P\left(Yf(X) > 0\right) < P_m\left(Yf(X) > \gamma_l\right) - \epsilon(\gamma_l, \gamma_l \delta) \right\}.$$

We claim the following.

Claim 42. For all $\gamma \in \Gamma_l$, $E(\gamma, \delta) \subseteq E_l$.

Proof of the Claim: For $\gamma \in \Gamma_l$ the statement $Yf(X) > \gamma$ implies $Yf(X) > \gamma_l$ so $P_m(Yf(X) > \gamma) \leq P_m(Yf(X) > \gamma_l)$. That $\epsilon(a, b)$ decreases with a implies $\epsilon(\gamma/2, b) \geq \epsilon(\gamma_l, b)$ for $\gamma \in \Gamma_l$. And because $\epsilon(a, b)$ decreases with b then $\epsilon(\gamma_l, (\gamma/2)\delta) \geq \epsilon(\gamma_l, \gamma_l\delta)$. It follows that $E(\gamma, \delta) \subseteq E_l$ for $\gamma \in \Gamma_l$. $\square$

Let

$$\epsilon(\gamma, \delta) := \sqrt{\frac{8}{m} \left(n \ln \left(\frac{16e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{2}{\delta} \right) \right)} \quad (72)$$

then

$$\begin{aligned} P^m(E_l) &= P^m \left(\exists f, P(Yf(X) \leq 0) > P_m(Yf(X) \leq \gamma_l) + \right. \\ &\quad \left. \sqrt{\frac{8}{m} \left(n \ln \left(\frac{16e^3 m^2 (4m-1)}{\gamma_l^2 n} \right) + \ln \left(\frac{2}{\gamma_l \delta} \right) \right)} \right) \\ &\leq \gamma_l \delta \end{aligned}$$

where the above inequality is obtained from Lemma 40. With Claim (42), it follows that the probability of the bad event (71) is bounded from above as follows

$$\begin{aligned} \sum_{l=1}^{\infty} P^m \left(\bigcup_{\gamma \in \Gamma_l} E(\gamma, \delta) \right) &\leq \sum_{l=1}^{\infty} P^m(E_l) \\ &\leq \delta \sum_{l=1}^{\infty} \left(\frac{1}{2} \right)^l \\ &\leq \delta. \end{aligned} \quad (73)$$

From (72) we have

$$\epsilon(\gamma/2, \gamma\delta/2) = \sqrt{\frac{8}{m} \left(n \ln \left(\frac{64e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{4}{\gamma\delta} \right) \right)}$$

and substituting this for $\epsilon(\gamma/2, \gamma\delta/2)$ in (70) means that the complement of the bad event is the event that (68) holds uniformly over all $f \in \mathcal{F}_n$ and all $\gamma > 0$, which from (73) has probability at least $1 - \delta$. $\square$

In the bound (68), as γ increases the empirical γ -margin error of f increases while the square root term decreases. Thus there is a tradeoff between the two terms of the sum on the right side of (68). Theorem 41 can be used as a basis for ‘model selection’ [11, Ch. 15] in order to choose a classifier $h \in \mathcal{H}_n$ which minimizes this sum with respect to γ . Its generalization error is bounded from above by an expression of the form

$$\inf_{\gamma \in (0,1]} \inf_{f \in \mathcal{F}_n} \left(P_m(Yf(X) \leq \gamma) + \sqrt{\frac{8}{m} \left(n \ln \left(\frac{64e^3 m^2 (4m-1)}{\gamma^2 n} \right) + \ln \left(\frac{4}{\gamma\delta} \right) \right)} \right)$$

8 Conclusions

This paper studies large-width classification on the ultrametric domain of p -adic integers $\mathbb{Z}_p$. The hypothesis class is defined by finite unions of p -adic balls, and the associated signed-width function represents each classifier as the sign of a real-valued function. This places the analysis within the framework of margin-based learning [10, 11] while retaining the explicit hierarchical geometry of the p -adic setting.

A central feature of the analysis is the strong structural regularity of balls in $\mathbb{Z}_p$: any two balls are either nested or disjoint. From this we obtain $\dim(\mathbb{Z}_p) = 1$ and, consequently, pseudo-dimension 1 for the associated class of distance functions. This leads to covering-number bounds for the class of signed-width functions and, in turn, to non-asymptotic generalization bounds for learning concepts that are unions of at most n balls. The bounds hold uniformly over the width parameter $\gamma \in (0, 1]$, so γ may be selected after observing a sample of finite size m . They quantify the tradeoff between empirical width error and a complexity term depending on n, m, γ , and the confidence parameter δ .

The analysis also gives a concrete interpretation of width in the p -adic setting. Since a ball is determined by a finite prefix in the p -adic expansion [27], larger width corresponds to a coarser ball and therefore to a simpler residue-class description of the concept. Thus, maximizing sample width favors classifiers that are simpler in a natural number-theoretic sense.

More broadly, the results suggest that ultrametric geometry can support a tractable and informative learning theory for hierarchical domains. In this perspective, the space $\mathbb{Z}_p$ serves not merely as a specialized number-theoretic object, but as a mathematically transparent instantiation of a more general framework for classification and generalization on hierarchical ultrametric spaces. Possible directions for further work include algorithmic aspects of learning over $\mathbb{Z}_p$, multi-category versions of the model, extensions to other ultrametric domains, and empirical evaluation on genuinely hierarchical data.

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